1. Reconstructed curve-seed selection window

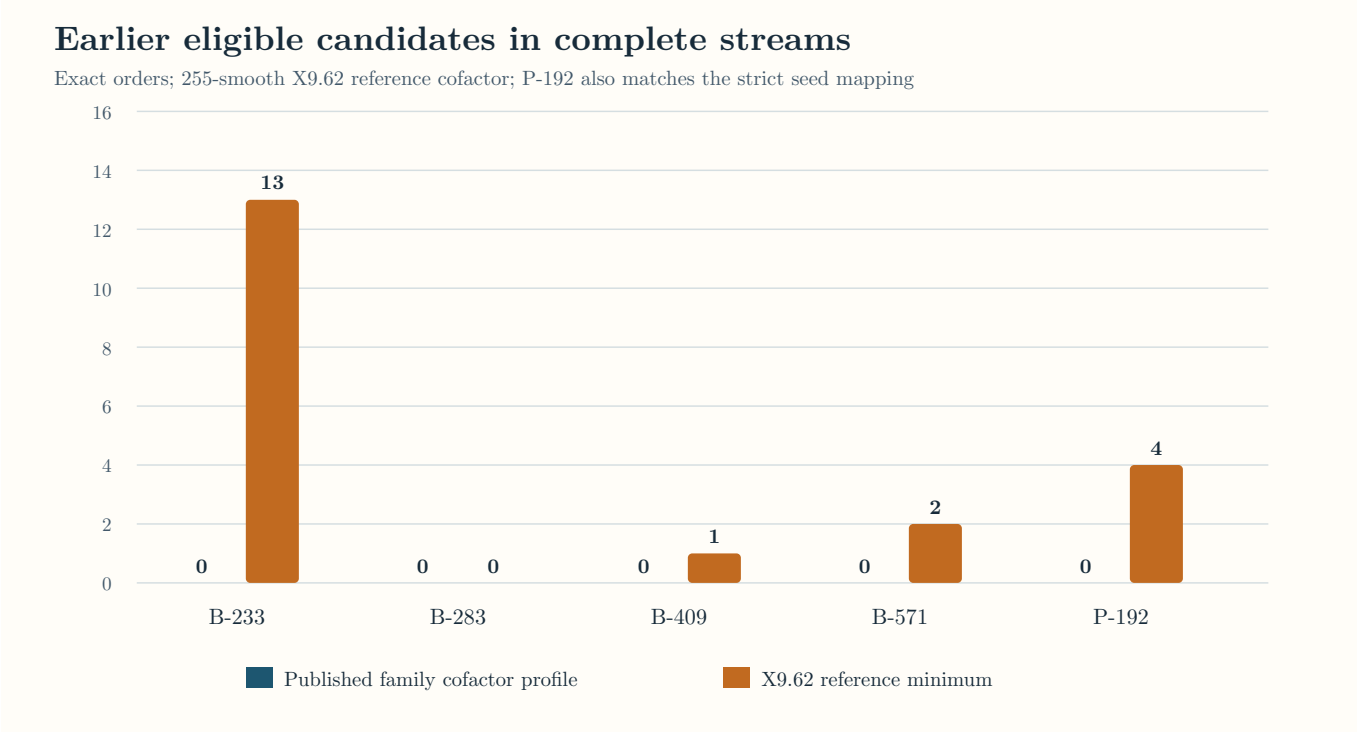
Evidence report · 16 October 2026 · Cross-checked six-stream census

The recovered `lrand48` coordinates regenerate 31,945 named target/index pairs, representing 22,073 distinct stream positions and including all fourteen published anchors. The host C library independently replayed every seed. Published-family-profile arithmetic is complete for the four binary streams and P-192. A merged six-stream prime-order census covers all 21,753 prime stream positions; our independent checker confirms its candidate-index set, accepted lists, 11 omitted orders, ten published orders, and 9,275 retained prefix rows.

1.1. What the completed streams show

Stream	Earlier seeds	Published cofactor profile	X9 reference subset
B-233	154	0	13
B-283	31	0	0
B-409	59	0	1
B-571	72	0	2
P-192	138	0	4

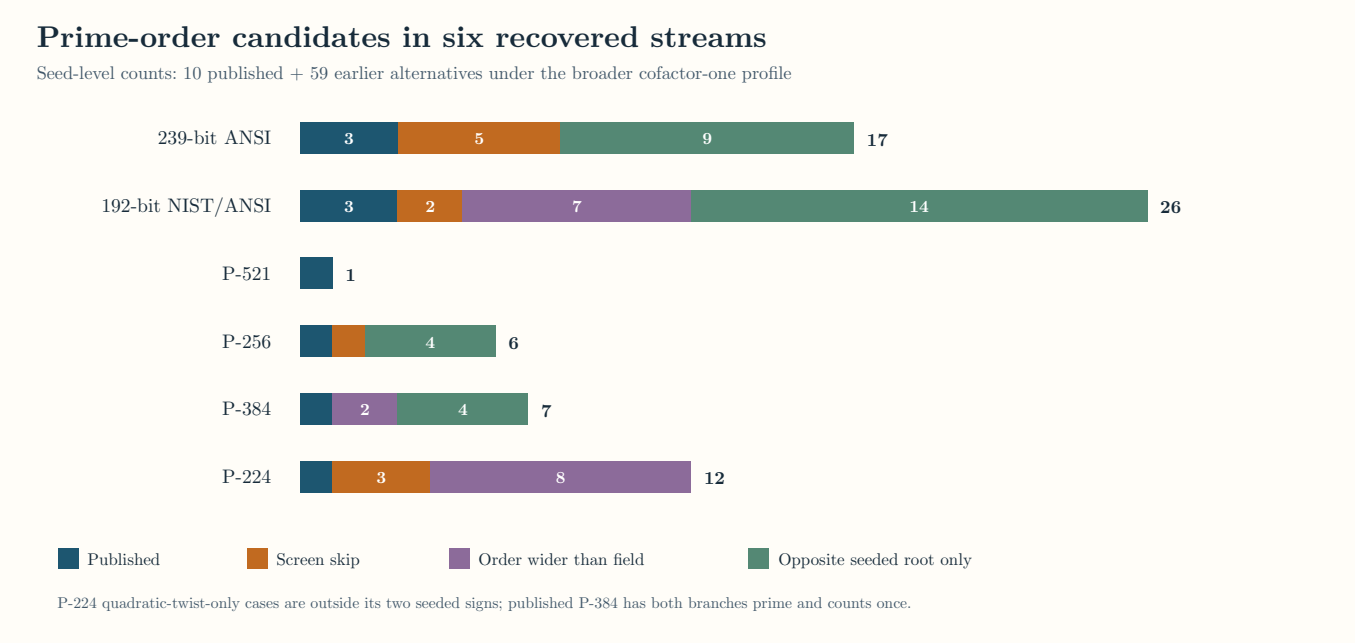
The X9 column uses a 255-smooth cofactor, the minimum subgroup bounds, MOV degrees 1–20, and the anomalous exclusion. The P-192 entry counts only published-compatible coefficients that also match the draft's strict seed formula; the distinct strict-formula curves still require their own counts.



1.2. Complete prime-order selection window

The retrospective selected-root screen gives 21 accepted entries: ten published and eleven unpublished. It requires prime group order, group-order bit length no greater than field bit length, nonanomalous order, and MOV nontriviality for degrees below 1024. Its width test is not a minimum rule found in the accessible September 1998 X9.62 draft; this is a reconstruction of a reported screen, not an authenticated 1997–1998 stopping rule.

The broader cofactor-one published-family profile admits either seeded root, uses the public minimum subgroup-size and MOV-20 tests, and applies no unsourced order-width cap. It gives 69 seed-level entries: ten published, eleven screen skips, seventeen wider prime-order cases, and 31 cases where only the opposite seeded root has prime order. P-224′s twist-only cases are outside its seeded sign pair. Published P-384 has prime order on both curve and twist; it is counted once. Independent affine-scalar and Hasse checks certify the claimed full group order on all 82 eligible root branches.

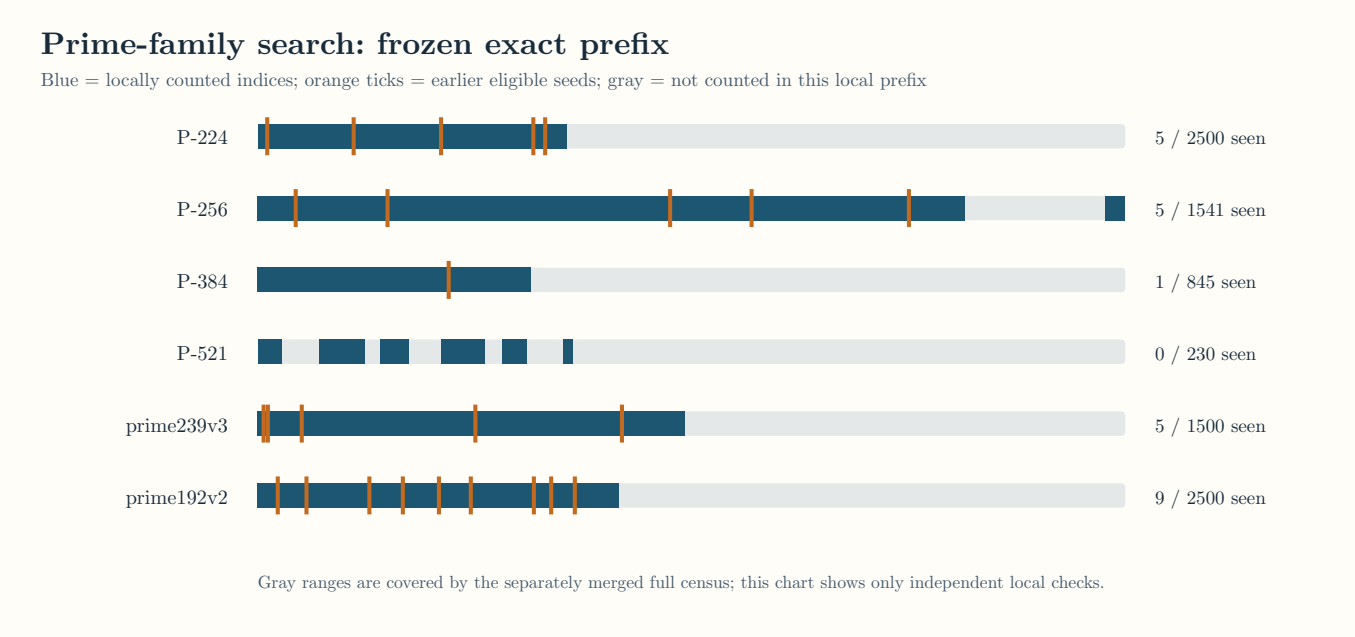


The eleven screen skips are 239-bit indices 36, 1562, 2491, 2682, 2863; 192-bit 2198, 5405; P-256 876; and P-224 777, 1485, 5736. Shared ANSI streams are counted once. Full candidate lists, field orders, and source hashes are in `raw/full_census_crosscheck_v4.json`.

1.3. Prime-family prefix: independent exact checks

The frozen pre-retry snapshot has exact results for the following earlier indices. These partial results agree with the completed imported census on their observed ranges. Shared ANSI streams are shown once; prime192v2 index 139 is the already published P-192 curve.

Representative stream	Earlier indices evaluated	Eligible seeds seen in prefix
P-224	2,500 / 7,025	5
P-256	1,541 / 1,841	5
P-384	845 / 2,687	1
P-521	230 / 1,135	0
prime239v3	1,500 / 3,047	5
prime192v2	2,500 / 6,012	9



All thirty accepted root branches in this snapshot have an independent affine scalar and Hasse-uniqueness certificate. The partial source files are retained with hashes; later completed and partial P-256 retry rows are bound in the campaign stop record and checked separately.

1.4. Specific earlier choices

P-256 index 876 is prime order and passes the tested published cofactor-one profile on the same exponentiation root as the published coefficient. The published index is 1842. The other four accepted P-256 indices seen so far (81, 276, 1049, 1383) use the opposite root. All nine published prime curves with field prime congruent to 3 modulo 4 use the exponentiation root; that is an observed pattern, not recovered constructor code.

P-224 index 1485 has a 224-bit prime group order larger than the published P-224 order. It satisfies both tested seed mappings, and the two coefficient roots are isomorphic over this field. All five earlier prime-order P-224 curves seen so far have orders larger than the published order. Thus a minimum subgroup-size threshold that admits published P-224 cannot exclude these earlier curves.

1.5. Meaning of the window

The eleven earlier selected-root alternatives falsify a simple first-eligible explanation under the displayed retrospective screen. A family cofactor preference explains the binary and P-192 skips in the completed streams. Neither the clock value nor the number of abandoned clock starts is authenticated. Private selection filters and the rule for choosing a coefficient root are unknown, so these counts do not identify intent or a hidden mathematical advantage.

1.6. Selection-window boundary

These results condition on each published field, the fixed coefficient a, and one reconstructed generator initialization. The recovered clock values are coordinates that reproduce the published seeds; they are not independent clock records. A constructor could have restarted, changed the clock input, or applied an undocumented preference before using this stream. The audit does not count those other possible searches.

For intact consecutive 20-draw seeds, the 48-bit LCG has at most `2^46` distinct candidate-start states; allowing arbitrary extra draws between intact windows raises the ceiling to `2^48`. These are recipe-specific possibility bounds, not historical attempt counts. A roughly 30-bit estimate requires a narrower timestamp window and a small number of chosen viable slots; it is a scenario rather than a universal bound.

Decision	Fixed by this audit	Still unverified
Seed stream	All positions up to each published index under the reconstructed initialization	Why that initialization or final index was chosen
Curve coefficient	Both seeded roots and the published SHA-1 mapping	The historical root convention and any rejected clock starts
Order and subgroup	Prime-order census; 9,275 overlapping prefix rows; completed binary and P-192 counts	Full X9 near-prime count and the private bounds or filters
Base point	Published points and independent subgroup-point certificates for observed survivors	The original base-point random state and selection procedure

An earlier publicly admissible curve shows available selection under the stated reconstruction. It does not prove that the constructor considered it. The absence of an earlier candidate in a completed stream closes only that stream and those filters. The prime-order census deliberately aborts on detected composite orders, so it cannot classify every near-prime curve allowed by a chosen X9 `lmax` and `rmin`. A hidden advantage would require its own mechanism, an executable route, subgroup preservation, and a costed comparison against known attacks; the seed audit alone supplies none of those.

The accessible X9.62 text is a September 1998 working draft; FIPS 186-2 was published in January 2000. Both postdate all fourteen reconstructed timestamps. The draft's prime seed formula disagrees with six of ten published prime anchors. Thus its rules are an explicit public reference, not an authenticated description of the original search.

1.7. Reproduction pointers

The complete raw candidate grid is `raw/candidates.jsonl`. Full prime-order classification, source hashes, and overlapping exact checks are in `raw/full_census_crosscheck_v4.json`; the imported raw point counts are in `../nist_seed_provenance_20261016/census/`. All 82 branch certificates are in `raw/full_prime_order_certificates.jsonl`. Binary and P-192 exact counts and certificates are in `raw/evaluations/` and linked from `REPORT.md`. The interrupted local prime campaign and stop/correction records remain in `raw/family_campaign/`. Source texts: X9.62 working draft NIST FIPS 186-2.